DHRUVA N

Aspiring Software Engineer | Computer Science & Business Systems Student
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PROFESSIONAL SUMMARY

Computer Science and Business Systems student seeking a Software Engineer role, with strong foundations in data structures, algorithms, object-oriented programming, and the software development life cycle (SDLC) in Java, Python, and C++. Built, tested, and deployed full-stack applications (React, Flask, FastAPI, REST APIs, PostgreSQL, SQL Server) on Render and Vercel, integrated LLM APIs into agentic AI workflows (LangGraph), and applied machine learning and computer vision (YOLOv8, OpenCV). Completed the HPE Software Engineering Job Simulation (Forage) with unit testing. Strong problem-solving, communication, and collaboration skills.

TECHNICAL SKILLS

- Programming Languages: Java, Python, C++, JavaScript, SQL
- Core CS & Practices: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Software Development Life Cycle (SDLC), Problem-Solving, Analytical Thinking, Troubleshooting
- Full-Stack & APIs: React.js, Redux Toolkit, HTML, CSS, Responsive Design, REST APIs, RESTful Web Services, API Integrations, Flask, FastAPI, Back-End Development
- Databases & Cloud: PostgreSQL, Microsoft SQL Server, SQL, DBMS, Cloud Deployment (Render, Vercel), Linux and Windows Environments
- Testing & Version Control: Unit Testing, Automated End-to-End Testing, Technical Documentation, Git, GitHub, Version Control
- Generative AI & ML: Generative AI, LLM Integration (Groq API), Prompt Engineering, LangGraph, Agentic Workflows, Machine Learning, Computer Vision, YOLOv8, OpenCV

PROJECTS

AIVOA — AI-Powered Customer Complaint Management System | *React, Redux Toolkit, FastAPI, LangGraph, Groq LLM, PostgreSQL*

- Designed, built, and deployed a full-stack complaint management system for pharmaceutical manufacturing, live on Vercel (frontend) and Render (backend + PostgreSQL).
- Designed a 7-node LangGraph agent workflow in Python — extraction, completeness checking, duplicate detection, risk classification, root cause analysis, CAPA recommendation, and summarization — powered by an LLM API (Groq), with prompts designed for each stage.
- Integrated the LLM through FastAPI REST APIs, a PostgreSQL database, and a responsive React + Redux Toolkit UI that auto-populates and risk-assesses records from uploaded documents.

PotholeAI — AI Pothole Detection & Repair Cost Estimation | *Python, Flask, YOLOv8, OpenCV, React, SQL Server, Leaflet*

- Built and deployed a full-stack ML platform on Render that detects potholes with YOLOv8, estimates size, depth, and volume with OpenCV, classifies severity, and forecasts repair cost.
- Designed a Flask REST API with 12 endpoints and a Microsoft SQL Server schema, with React pages (interactive Leaflet map, reports feed, admin dashboard); wrote an automated end-to-end test suite and project documentation (README).

AI-Driven Face Recognition-Based Attendance Management System | *Python, OpenCV, Machine Learning, Database Management*

- Engineered an automated attendance system using computer vision and ML, replacing manual roll-call with real-time facial recognition that logs individuals directly into a database, tuning detection to handle varied lighting and angles.

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science and Business Systems | Malnad College of Engineering | 2023 – 2027
PUC / Class 12 — Creative Pre University College, Karkala — 83%
SSLC / Class 10 — Amritha Bharathi Vidya Kendra, Hebri

CERTIFICATIONS

- Artificial Intelligence in 3 Hours — Infosys Springboard (May 2026)
- Software Engineering Job Simulation — Hewlett Packard Enterprise (HPE) via Forage (October 2026)
- Certificate of Participation, Round 2: Detailed Presentation Submission — EY Techathon 6.0

ACHIEVEMENTS & LEADERSHIP

- Advanced to the second round of EY Techathon 6.0, a national-level innovation competition.
- Coordinated college-level technical events, demonstrating team leadership and project coordination.

LANGUAGES

English (Fluent) | Kannada (Fluent) | Hindi (Intermediate) | Spanish (Beginner)